

AACDB: Antigen-Antibody Complex Database — a Comprehensive Database Unlocking Insights into Interaction Interface

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eLife Assessment

This **useful** manuscript provides a newly curated database (termed AACDB) of antibody-antigens structural information, alongside annotations that are either taken and from the PDB, or added de-novo. Sequences, structures, and annotations can be easily downloaded from the AACDB website, speeding up the development of structure-based algorithms and analysis pipelines to characterize antibody-antigen interactions. The methodology presented for this data curation is **solid**. The curated dataset will be of broad interest and value to researchers interested in antibody-antigen interactions.

<https://doi.org/10.7554/eLife.104934.2.sa3>

Abstract

Antibodies are critical components of the vertebrate immune system and possess a wide array of biomedical applications. Elucidating the complex interactions between antibodies and antigens is an important step in drug development. However, the complex and vast nature of the data presents significant challenges in accurately identifying and comprehending these interactions. To overcome these challenges and deepen our understanding of the antibody-antigen interface, we developed the Antigen-Antibody Complex Database (AACDB). The current version provides a comprehensive collection of 7,498 manually processed antigen-antibody complexes, ensuring accuracy and detail. This database provides extensive metadata and rectifies annotation errors found in the PDB database. Furthermore, it integrates data on antibody developability and antigen-drug target relationships, making it valuable for assisting new antibody therapies development. Notably, the database includes comprehensive paratope and epitope annotation information, thereby serving as a valuable benchmark for immunoinformatics research. The AACDB interface is designed to be user-friendly, providing researchers with powerful search and visualization tools that enable effortless querying, manipulation, and visualization of complex data.

Researchers can access AACDB completely online at <http://i.uestc.edu.cn/AACDB>. Regular updates are promised to ensure the timely provision of scientific and valuable information.

1. Introduction

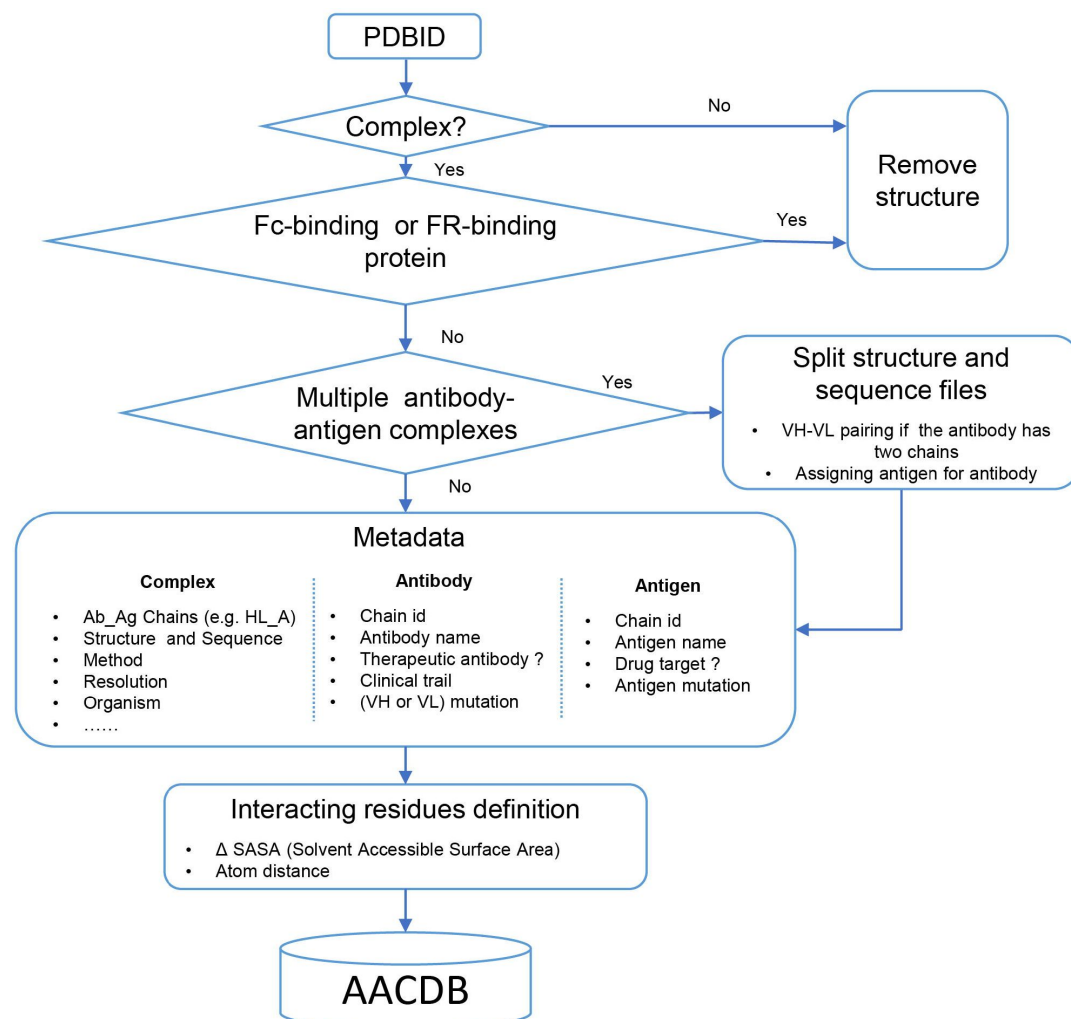
Antibodies are not only an essential part of the vertebrate immune system, but also has wide application in biomedical field [1, 2]. In clinical practice, the advent of monoclonal antibodies (mAbs) has transformed the treatment for various cancers, autoimmune diseases, and numerous other conditions. Due to their large binding interfaces, high affinity and specificity, antibody drugs have successfully targeted numerous proteins that were previously impervious to small molecule therapies. The expansion of antibody drug formats, such as antibody-drug conjugates (ADCs), domain antibodies, bispecific antibodies and antibody fusion proteins, has broadened their indications to a wider range of diseases including blood disorders, infections, neurological conditions, ocular diseases, and metabolic disorders [3-5]. Antibody drug development typically encompasses three stages: preclinical research, clinical trial and post-marketing surveillance study [6]. The preclinical research of antibodies mainly refers to a series of in vitro experiments and animal model studies conducted before the application of antibodies in clinical trials. In this stage, an important task is to determine the binding interfaces and interaction sites between antibodies and their drug targets. Understanding how antibodies specifically interact with their targets or antigens can enable help optimize antibody drug leads further, as well as provide insights into mechanisms of interaction [7].

The field of machine learning and deep learning-based methods for antigen and antibody binding interfaces prediction have gained significant attention in recent years [8-17]. The advancement of this field heavily relies on a reliable foundation of data. Although the Protein Data Bank (PDB) [18, 19] is a well-established repository for protein complexes, identifying antigen-antibody complexes within its extensive collection of general protein structures presents considerable challenges. In 2018, AbDb [20] was introduced as a specialized database for antigen-antibody complexes. However, its adoption has been limited by the absence of advanced visualization tools, interactive functionalities and abundant metadata annotations. We acknowledge that AbDb has made exceptional contributions to the field by aggregating antibody-antigen structures over the past decade. Our work builds upon these foundations, aiming to address complementary challenges in annotation and interoperability. In comparison, the series of SAbDab databases [21-23] indeed provide a comprehensive data source of antibody structure and achieve in time updating. However, this rapid updating process may inadvertently overlook a significant amount of information that requires thorough verification, thereby increasing the likelihood of incorporating erroneous annotations directly from the PDB database.

A further challenge in the field of interface prediction is the lack of a standardized definition for interacting amino acids. Some datasets are based on differences in the solvent accessible surface area (ASASA) for each residue upon binding [9, 24]; Others are based on distances between atoms [8, 13-15, 25]. Different studies employ different datasets and definitions, resulting in a lack of comparability. There is an urgent need for a standard benchmark dataset to enable objective comparisons of these methods. None of the databases mentioned above provide details of interaction residues. To address these limitations, we developed the Antigen-Antibody Complex Database (AACDB). It has a user-friendly interface for convenient querying, manipulation, browsing, and visualization of comprehensive information about antibody-antigen complexes. Notably, AACDB also provides detailed information on interacting residues using two methods (ASASA and atom distance), facilitating the benchmarking studies of the predictive methods for paratopes and epitopes.

2. Materials and methods

We search the PDB database using the keywords “antibody”, “antigen”, “immunoglobulin” and “complex”. Over 32000 experimental structures were extracted (up to December 2023). Each entry was processed following the steps in [Figure 1](#).

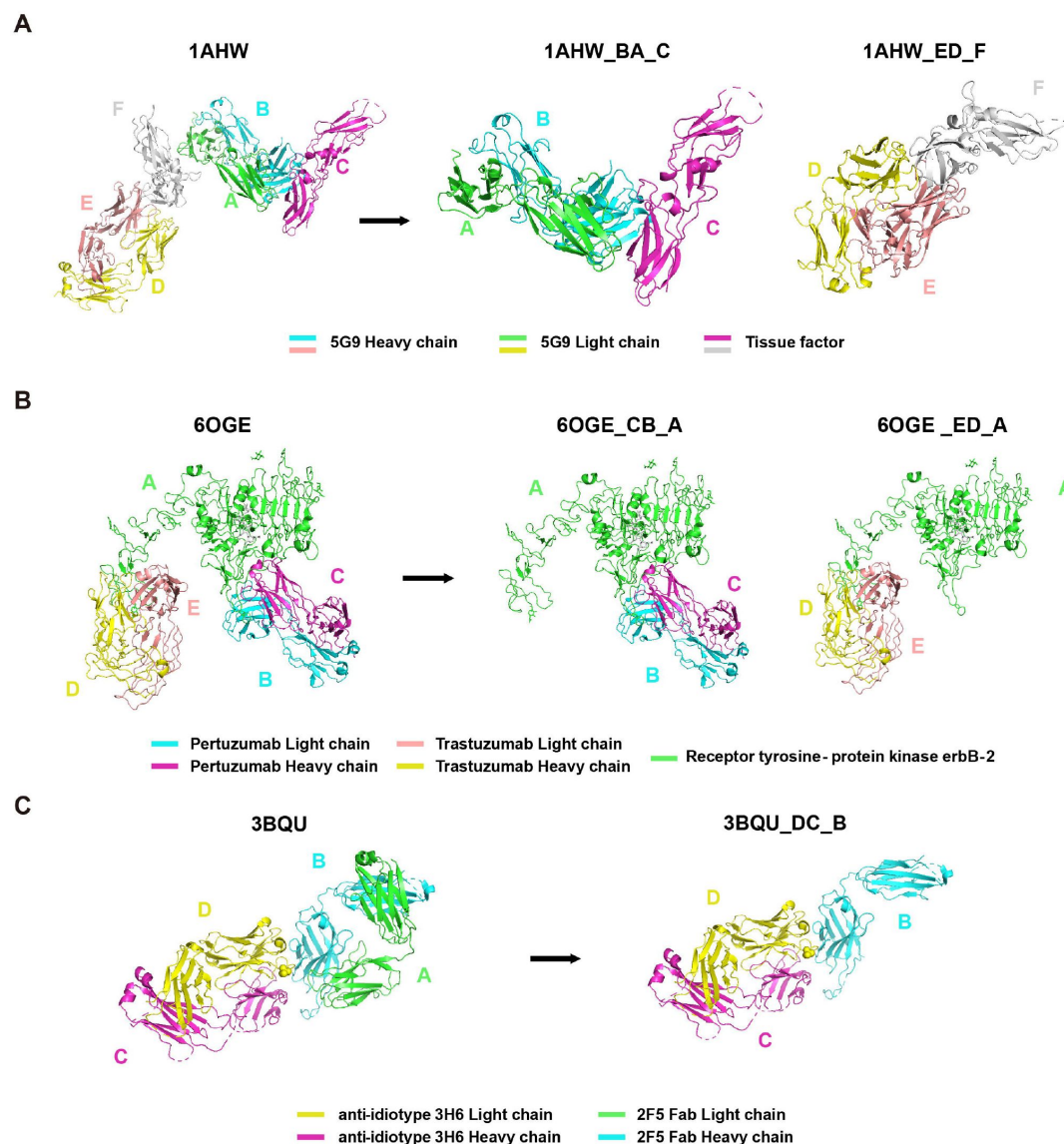


2.1 Entry screening

We aimed to retrieve all antibody-related complexes from the PDB using the chosen search approach. However, we encountered a notable number of false hits in the results, prompting us to implement a stringent structural filtering process. Antibody complexes were defined as structures containing at least one antibody molecule and another protein exceeding 50 amino acids in length. Structures that lacked antibodies or consisted solely of one type of antibody were excluded from further analysis. It is crucial to emphasize that not all proteins that bind to antibodies qualify as antigens. For instance, immunoglobulin-binding proteins like Protein A and Protein G, expressed by *Staphylococcus aureus* and *Streptococcal* species, are commonly employed for antibody purification procedures [26]. Although protein A may bind to the Fab region of antibodies, the interacting amino acids might be located in the framework region (FR) rather than the complementarity-determining regions (CDRs) [27]. These proteins are often misidentified as antigen. Consequently, structures exhibiting such characteristics require meticulous verification and should be excluded from our dataset.

2.2 PDB splitting

We directly incorporated the PDB entry of antigen-antibody complex with only one antigen and one antibody into the AACDB database. However, quite a few antigen-antibody complexes contain several antigens and antibodies. In such cases, before splitting the structures, we need to determine the correct pairing of light and heavy chains by examining the information on equivalent chain interactions in the PDBsum database [28]. Next, we assign the correct antigen chains to the antibodies by identifying whether the remaining chains in the structure interact with the CDR regions of determined antibodies. For example, in the case of 1AHW, which contains two copies of antigen-antibody complexes, chains BA and DE are identified as two antibody pairs, where chains C and F bind to BA and DE, respectively. Consequently, 1AHW is split into two files, BAC and DEF (**Figure 2A**). In 6OGE, Pertuzumab (chains C and B) and Trastuzumab (chains E and D) bind to different epitopes of the receptor tyrosine-protein kinase erbB-2 (chain A). This generates two records in AACDB (CBA and EDA) (**Figure 2B**). Specifically, In the analysis of anti-idiotypic antibody complexes, the partner antibody is treated as a dual-chain antigen, necessitating individual evaluation of heavy chain and light chain interactions with the anti-idiotypic component. When both chains engage in binding, the complex is divided into two distinct entries: “anti-idiotypic antibody-heavy chain” and “anti-idiotypic antibody-light chain.” Conversely, if only one chain participates, the non-interacting chain will be excluded from AACDB records. For instance, in 3BQU, the anti-idiotypic 3H6 Fab (chains DC) only interact with the heavy chain (chain B) of 2F5 Fab. In AACDB, 2F5 Fab will be split and the light chain (chain A) will be discarded(**Figure 2C**).



For each PDB entry, we utilized the corresponding split chains information to divide the downloaded .pdb based on the ATOM records. Notably, certain structures are exclusively provided in the “mmCIF” format by the PDB database. While “.pdb” and “.cif” files store atomic coordinates in distinct text formats, the segmentation of these structure files is automatically performed based on manually annotated antibody-antigen chains. To address this, we integrated these considerations into our file processing pipeline enabling a fully automated file segmentation process. Additionally, the Naccess software does not support .cif files as input, we converted the “.cif” files into “.pdb” format files while performing the splitting process (see **Interacting residues definition**). Furthermore, we refined the annotation of the split “.fasta” files to ensure coherence with AACDB records.

2.3 Metadata

AACDB provides detailed metadata for each entry, including chain IDs, antibody name, antigen name, method, resolution, organism, and more. To ensure data accuracy, we have conducted comprehensive verification by consulting original literature sources. AACDB has addressed many annotation errors identified within the corresponding PDB entries. These errors include but not limited to: (1) mislabeling of species (e.g., the entry 7WRL where the organism of BD55-1239H antibody was erroneously labeled as “SARS coronavirus B012”); (2) Resolution annotation errors

(e.g., 1NSN, in which the resolution of 2.9 Å is misannotated as 2.8 Å); (3) mislabeling of antibody chains as other proteins (e.g., in 3KS0, the light chain of B2B4 antibody was misnamed as heme domain of flavocytochrome b2); (4) misidentification of heavy chains as light chains (e.g., both two chains of antibody were labeled as light chain in 5EBW); (5) mutation status annotation errors. We have identified cases in which PDB entries indicate “NO” for mutations, while in reality, mutations exist (e.g., bevacizumab (Avastin) in 6BFT was labeled as none mutation. When aligned with the bevacizumab sequence, however, mutation T8D/T30D in heavy chain and S52D/S53D in light chain were observed.); and (6) incomplete annotations. Certain entries only provide the name of the mutant without specifying the precise mutation site (e.g., in 7SU1, antibody was described as Ipilimumab variant Ipi.106. but PDB database do not provide any mutation amino acid or position, which can be identified by blast to Ipilimumab sequence).

We carefully checked each entry manually to find out all possible annotation problems. Given the context-dependent nature of mutations, sequence alignment was not universally performed across all antibody sequences. Instead, therapeutic antibody sequences served as definitive references. For example, for entries in the PDB labeled as “Bevacizumab mutant” or “Ipilimumab variant Ipi.106” where the “Mutation(s)” field was annotated as “NO,” we retrieved corresponding therapeutic antibody sequences from Thera-SAbDab and performed sequence alignment with the PDB entries to identify mutated residues. For other antibodies lacking well-defined wild-type references, mutation information was directly extracted from the PDB or original literature. All corrections have been publicly archived in AACDB.

The antibody nomenclature follows the title of the corresponding search entry in the RCSB PDB database, with verification done through the original literature. In cases where the name in the original literature differs from that in the RCSB PDB, we used the name in the published literature as the standard. Furthermore, for antibody fragments lacking names in both the RCSB PDB database and original literature, we adopt a naming convention of “PDBID” + “antibody fragment” (e.g., 4WEB Fab).

Biological and physicochemical properties are critical considerations in the development pipelines for therapeutic antibodies. These properties include solubility, immunogenicity, aggregation tendencies, expression level, stability, and hydrophobicity. We provide the International Non-proprietary Name (INN) and clinical trial information for each therapeutic antibody entry, linked to the DOTAD database [29]. Numerous antigens have been successfully identified as targets for antibodies or small molecule drugs. We conducted a comparison between antigen sequences and the drug targets listed in the DrugBank database [30]. A threshold of percent identity > 90% was applied to determine the corresponding drug targets.

2.4 Interacting residues definition

We labeled the interacting residues based on SASA and atom distance. Naccess V2.1.1 and Bio.PDB module were employed to calculate SASA value for each residue in antibody and antigen, respectively. The residues with a SASA loss (Δ SASA) in binding of more than $1\text{\AA}^2$ were classified as interacting residues. In addition, we also defined another set of interacting paratope-epitope residues by a distance cutoff of $6\text{\AA}$. Two amino acids are considered as interacting residues if they have at least one pair of non-hydrogen atoms atom within a distance of $6\text{\AA}$.

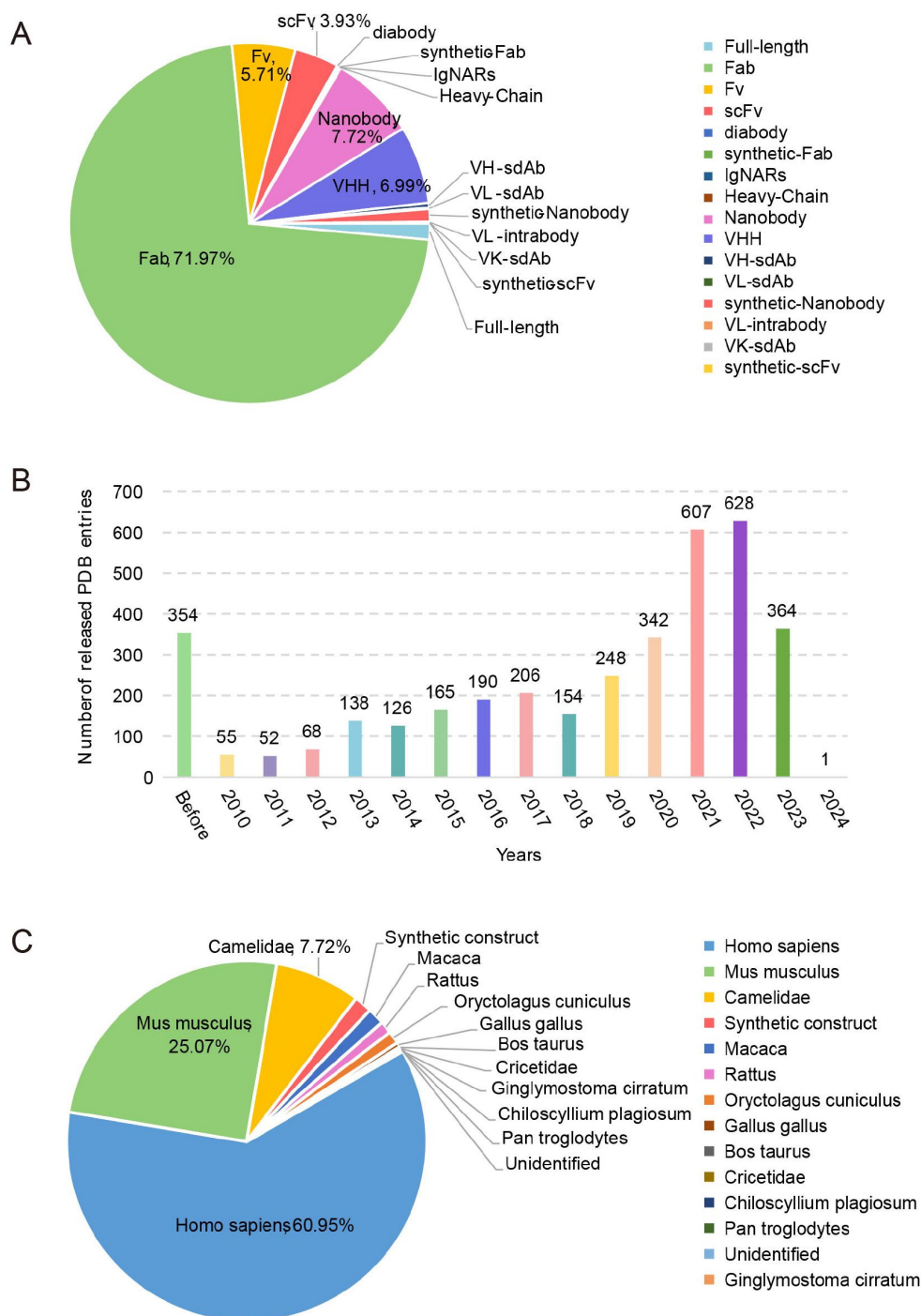
2.5 Data integration and website implementation

The main data processing algorithm is implemented in Python. The front-end web interface of AACDB was constructed by HTML and enhanced with JavaScript, CSS and Bootstrap technologies. We developed a dynamic 3D structure visualization window based on PV, a WebGL-based protein viewer, inspired by Dunbar *et al.* [21]. All the data were managed within the MySQL database system. For the back-end functionality, PHP is utilized to enable data browsing, searching, and downloading features.

3. Results

3.1 Statistics

From over 32000 experimental structures in the PDB database, we manually curated 7498 antigen-antibody entries for the current version of AACDB, encompassing to 16 antibody fragment types across 14 species(**Figure 3** [↗](#)). It is obvious that fab fragments and human antibodies accounted for the largest proportion of the data, accounting for 71.98% and 60.95%, respectively. Our statistical analysis reveals a significant increase in the number of antigen-antibody complex entries within the PDB during the period from 2021 to 2023. These entries accounted for approximately 45% of the total antigen-antibody complex entries. Our search was conducted with an end date of November 2023. Notably, **Figure 2B** [↗](#) contains a 2024-dated entry resulting from database versioning updates: the original 7SIX record (added November 16, 2022) was superseded by the 8TM1 entry through an official revision on January 12, 2024. Furthermore, the developability properties of antibodies in 325 entries can be queried in the DOTAD database, at the meanwhile, 3,733 antigen records have been identified as drug targets in DrugBank (data not shown).



3.2 Database browse and search

All the data can be browsed directly by clicking the “Datasets” item on the top menu ([Figure 4](#)). The summary table includes nine columns as following:

Datasets

Search

PDBID: Antibody: Protein:
Organism: Resolution(<): All fields:

[Submit search](#) [Search](#) [Click to download the search results](#)

Results list: 117 hits (Download the results: [txt](#) or [csv](#)) [Click to get original literature](#)

[Click for the details](#)

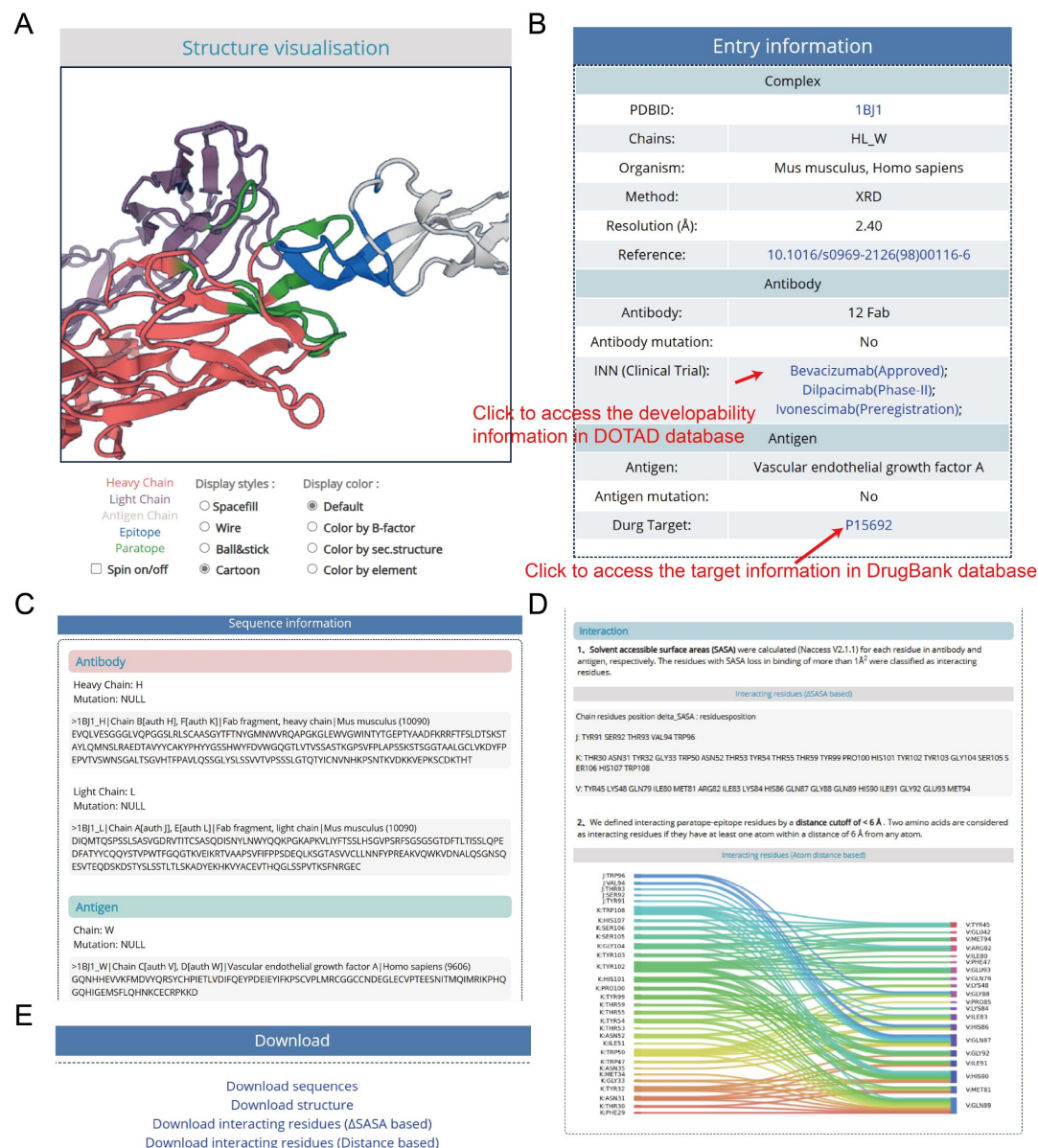
AACDB_ID	PDBID	Chains	Antibody	Protein	Organism	Method	Resolution	Reference
1	1A2Y	BA_C	D1.3 Fv	Lysozyme C	Mus musculus, Gallus gallus	XRD	1.50	10.1021/bi980148j
2	1BQL	BA_Y	HYHEL-5 Fab	BOBWHITE QUAIL Lysozyme	Mus musculus, Colinus virginianus	XRD	2.60	10.1002/(SICI)1097-0134(199609)26:1<55: AID-PROT5>3.0.CO;2-F
3	1BVK	BA_C	HULYS11 Fv	Lysozyme	Homo sapiens, Gallus gallus	XRD	2.70	10.1084/jem.187.4.479
4	1BVK	ED_F	HULYS11 Fv	Lysozyme	Homo sapiens, Gallus gallus	XRD	2.70	10.1084/jem.187.4.479
5	1C08	BA_C	HYHEL-10 Fv	Hen Egg White Lysozyme	Mus musculus, Gallus gallus	XRD	2.30	10.1074/jbc.274.39.27623
6	1DQJ	BA_C	HyHEL-63 Fab	Lysozyme C	Mus musculus, Gallus gallus	XRD	2.00	10.1021/bi000054i
7	1DZB	A_X	1F9 scFv	Turkey Egg White Lysozyme	Mus musculus, Meleagris gallopavo	XRD	2.00	10.1006/jmbi.2000.3971
8	1DZB	B_Y	1F9 scFv	Turkey Egg White Lysozyme	Mus musculus, Meleagris gallopavo	XRD	2.00	10.1006/jmbi.2000.3971
9	1FBI	HL_X	F9.13.7 Fab	GUINEA FOWL Lysozyme	Mus musculus, Numida meleagris	XRD	3.00	10.1074/jbc.270.30.18067
10	1FBI	QP_Y	F9.13.7 Fab	GUINEA FOWL	Mus musculus, Numida	XRD	3.00	10.1074/jbc.270.30.18067

1. AACDB_ID: The unique id in the AACDB database, linking to the “Detail” page;
2. PDBID: The identifier of the RCSB PDB database (<https://www.rcsb.org/>), linking to PDB;
3. Chains: The chain ids containing in this entry. Antibody and antigen chains were separated by “_”. The heavy chain id precedes the light chain id if antibody is complete;
4. Antibody: This column is represented by “antibody name + fragment”;
5. Protein: The corresponding antigen;
6. Organism: The source organism of antibody and antigen;
7. Method: The experiment method used to solve the structure;
8. Resolution: The indicator that measures the resolution of protein structures in experiments is expressed in units of angstroms (Å).
9. Reference: The DOI linker of original literature that produced this structure.

The table in AACDB can be easily searched through the search panel at the top section of the “Datasets” page. Users can perform a quick search by specifying one or more fields and entering relevant keywords. The search results can be downloaded as files in either txt or csv format. **Figure 4** shows the search results using the keyword “Lysozyme” in “Protein” column, returning 117 hits.

An individual structure can be accessed using its AACDB_ID accession code. When click the AACDB_ID hyperlink, the user will be brought to its details page as shown in **Figure 5**. There, the complex structure can be visualized with different colors and styles (**Figure 5A**). Besides the visualization window, we provide more details of this entry, including the mutation, INN and clinical trial of antibody, the ID in DrugBank of antigen (if exist) (**Figure 5B**). Under the structure information tab, further details about each chain can be found. These include: sequence, mutate amino acid type and position, interacting residues in each chain based on the ΔSASA method, interacting plot of paratope-epitope residues by a distance cutoff of < 6Å (**Figure 5D**). It should be pointed out that while our manuscript emphasizes widely accepted thresholds for

consistency with prior benchmarks, AACDB explicitly provides raw Δ SASA and distance values for all annotated residues. Users can dynamically filter the data from the downloaded file (e.g., selecting 5 Å instead of 6 Å) by excluding entries exceeding their preferred thresholds.



3.3 Data download

We provide two ways for downloading the data:

1. Download data of the single entry.

When access the detailed information about an entry using the corresponding AACDB_ID, user can click the hyperlink at the “Download” section of the bottom of the page to download the data for a single certain entry (Figure 5E).

2. Download all the data of AACDB.

AACDB provide the download page for users. All the sequence and structure files and the interacting data based on different methods were packaged in different .zip file that can be downloaded.

Moreover, the website provides a user-friendly 'Help' page that presents a step-by-step tutorial to assist users in manipulating, querying, browsing and downloading the AACDB database.

4. Discussion and conclusion

Research on antigen-antibody interactions contributes to the advancement of the antibody-related industry. Databases such as PDB and SabDab provide the foundational data for this purpose. However, there are still many unresolved issues. In this study, we developed the AACDB database to provide a curated and reliable dataset of antigen-antibody interactions. During the process of data collection and organization, we identified numerous annotation errors in the PDB database. Some of these errors had been directly introduced into SabDab. For example, the species of the antibody in 7WRL was incorrectly labeled as “SARS coronavirus B012” in both PDB and SabDab. We also invested significant effort and time to manually cross-reference with original literature in order to rectify these errors and exclude antibody binding proteins that were erroneously annotated as antigens by SabDab. Apart from the curation and reannotation of structural data, AACDB offers features not provided by other antigen-antibody complex databases: 1) AACDB's data processing pipeline supports mmCIF files, and 2) we provide amino acids in the interaction interface through two methods, enabling the definition of unified standards for epitopes and paratopes. This provides a more accurate and comprehensive benchmark dataset for developed interaction interface prediction tools, enhancing the comparability of various tools.

However, AACDB still has some limitations. Despite our best efforts, the limitations of our team's resources and knowledge mean that our database may not capture all antigen-antibody complex structures. Since the data is manually curated, eliminating errors completely during the information processing is a challenge. While we strive to fill in any gaps, we also hope that experts and users within the community can provide timely feedback to help us improve these issues. Additionally, currently AACDB only includes antigen proteins with a length greater than 50 amino acids. The current AACDB release exclusively includes antigen proteins exceeding 50 amino acids in length. Future iterations will expand structural coverage to diverse antigen classes (peptides, nucleic acids, haptens) while incorporating literature-validated affinity data through systematic cross-referencing.

In summary, AACDB is a novel database of antigen-antibody complexes that provides information on antibody-developability, antigen-drug target relationships, and detailed antigen-antibody interaction interfaces. It is fully accessible at <http://i.uestc.edu.cn/AACDB>. We are committed to a regular data update, ensuring that researchers in immunoinformatics have access to timely and valuable resources.

Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant numbers 62071099, 62371112] and Sichuan Province Science and Technology Support Program [2024NSFSC0636]. We are grateful to Steve's Scholar Memory Team for polishing our final manuscript.

Additional information

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Reviewer #1 (Public review):

This work introduces and describes a useful curation pipeline of antibody-antigen structures downloaded from the PDB database. The antibody-antigen structures are presented in a new database called AACDB - with associated website - alongside annotations that were either corrected from those present in the PDB database, or added de-novo with solid methodology. Sequences, structures and annotations can be very easily downloaded from the AACDB website, speeding up the development of structure-based algorithms and analysis pipelines to characterize antibody-antigen interactions. However, AACDB is missing some important annotations that I believe would greatly enhance its usefulness, such as binding affinity annotations.

I think the potentially most significant contribution of this database is the manual data curation to fix errors present in the PDB entries, by cross-referencing with the literature. The authors also seem to describe, whenever possible, the procedures they took to correct the annotations.

I have personally verified some of the examples presented by the authors, and found that SAbDab appears to fix the mistakes related to mis-identification of antibody chains, but not other annotations.

"(1) the species of the antibody in 7WRL was incorrectly labeled as "SARS coronavirus B012" in both PDB and SabDab" → I have verified the mistake and fix, and that SAbDab does not fix is, just uses the pdb annotation.

"(2) 1NSN, the resolution should be 2.9 , but it was incorrectly labeled as 2.8" → I have verified the mistake and fix, and that saabdab does not fix it, just uses the PDB annotation.

"(3) mislabeling of antibody chains as other proteins (e.g. in 3KS0, the light chain of B2B4 antibody was misnamed as heme domain of flavocytochrome b2)" → SAbDab fixes this as well in this case.

"(4) misidentification of heavy chains as light chains (e.g. both two chains of antibody were labeled as light chain in 5EBW)" → SAbDab fixes this as well in this case.

I believe the splitting of the pdb files is a valuable contribution as it standardizes the distribution of antibody-antigen complexes. Indeed, there is great heterogeneity in how many copies of the same structure are present in the structure uploaded to the PDB, generating potential artifacts for machine learning applications to pick up on. That being said, I have two thoughts both for the authors and the broader community. First, in the case of multiple antibodies binding to different epitopes on the same antigen, one should not ignore the potentially stabilizing effect that the binding of one antibody has on the complex, thereby enabling the binding of the second antibody. In general, I urge the community to think about what is the most appropriate spatial context to consider when modeling the stability of

interactions from crystal structure data. Second, and in a similar vein, some antigens occur naturally as homomultimers - e.g. influenza hemagglutinin is a homotrimer. Therefore, to analyze the stability of a full-antigen-antibody structure, I believe it would be necessary to consider the full homo-trimer, whereas in the current curation of AACDB with the proposed data splitting, only the monomers are present.

I think the annotation of interface residues is a very useful addition to structural datasets.

I am, however, not convinced of the utility of *change* in SASA as a useful metric for identifying interacting residues, beyond what is already identified via pairwise distances between the antibody and antigen residues. If we had access to the unbound conformation of most antibodies and antigens, then we could analyze the differences in structural conformations upon binding, which can be in part quantified by change in SASA. However, as only bound structures are usually available, one is usually forced to approximate a protein's unbound structure by computationally removing its binding partner - as it seems to me the authors of this work are doing.

Some obvious limitations of AACDB in its current form include:

AACDB only contains entries with protein-based antigens of at most 50 amino-acids in length. This excludes non-protein-based antigens, such as carbohydrate- and nucleotide-based, as well as short peptide antigens.

AACDB does not include annotations of binding affinity, which are present in SAbDab and have been proven useful both for characterizing drivers of antibody-antigen interactions (cite <https://www.sciencedirect.com/science/article/pii/S0969212624004362?via%3Dihub>) and for benchmarking antigen-specific antibody-design algorithms (cite <https://www.biorxiv.org/content/10.1101/2023.12.10.570461v1>)

<https://doi.org/10.7554/eLife.104934.2.sa2>

Reviewer #2 (Public review):

Summary:

Antibodies, thanks to their high binding affinity and specificity to cognate protein targets, are increasingly used as research and therapeutic tools. In this work, Zhou et al. have created, curated and made publicly available a new database of antibody-antigen complexes to support research in the field of antibody modelling, development and engineering.

Strengths:

The authors have performed a manual curation of antibody-antigen complexes from the Protein Data Bank, rectifying annotation errors; they have added two methods to estimate paratope-epitope interfaces; they have produced a web interface capable of effective visualisation and of summarising the key useful information in one page. The database is also cross-linked to other databases that contain information relevant to antibody developability and therapeutic applications.

Weaknesses:

The database does not import all the experimental information from PDB and contains only complexes with large protein targets.

Comments on revisions: I thank the authors for having incorporated my feedback and I look forward to the next releases of this database.

<https://doi.org/10.7554/eLife.104934.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public reviews:

Reviewer #1:

(1) This manuscript introduces a useful curation pipeline of antibody-antigen structures downloaded from the PDB database. The antibody-antigen structures are presented in a new database called AACDB, alongside annotations that were either corrected from those present in the PDB database or added de-novo with a solid methodology. Sequences, structures, and annotations can be very easily downloaded from the AACDB website, speeding up the development of structure-based algorithms and analysis pipelines to characterize antibody-antigen interactions. However, AACDB is missing some key annotations that would greatly enhance its usefulness.

Here are detailed comments regarding the three strengths above:

I think potentially the most significant contribution of this database is the manual data curation to fix errors present in the PDB entries, by cross-referencing with the literature. However, as a reviewer, validating the extent and the impact of these corrections is hard, since the authors only provided a few anecdotal examples in their manuscript.

I have personally verified some of the examples presented by the authors and found that SAbDab appears to fix the mistakes related to the misidentification of antibody chains, but not other annotations.

(a) "the species of the antibody in 7WRL was incorrectly labeled as "SARS coronavirus B012" in both PDB and SabDab" → I have verified the mistake and fix, and that SAbDab does not fix is, just uses the pdb annotation.

(b) "1NSN, the resolution should be 2.9 , but it was incorrectly labeled as 2.8" → I have verified the mistake and fix, and that sabdab does not fix it, just uses the PDB annotation.

(c) "mislabeling of antibody chains as other proteins (e.g. in 3KS0, the light chain of B2B4 antibody was misnamed as heme domain of flavocytochrome b2)" → SAbDab fixes this as well in this case.

(d) "misidentification of heavy chains as light chains (e.g. both two chains of antibody were labeled as light chain in 5EBW)" → SAbDab fixes this as well in this case.

I personally believe the authors should make public the corrections made, and describe the procedures - if systematic - to identify and correct the mistakes. For example, what was the exact procedure (e.g. where were sequences found, how were the sequences aligned, etc.) to find mutations? Was the procedure run on every entry?

We appreciate the reviewer's valuable feedback. Our correction procedures combined manual curation with systematic sequence analysis. While most metadata discrepancies were resolved through cross-referencing original literature, we implemented a structured approach for identifying mutations in specific cases. For PDB entries labeled as variants (e.g., "Bevacizumab mutant" or "Ipilimumab variant Ipi.106") where the "Mutation(s)" field was annotated as "NO," we retrieved the canonical therapeutic antibody sequence from TheraSAbDab, then performed pairwise sequence alignment against the PDB entry using BLAST program to identified mutated residues.

This procedure was not applied to all entries, as mutations are context-dependent. Therapeutic antibodies have well-defined reference sequences, enabling systematic alignment. For antibodies lacking unambiguous wild-type references (e.g., research-grade or non-therapeutic antibodies), mutation annotations were directly inherited from the PDB or literature.

All corrections have been publicly archived in AACDB. We have added a detailed discussion of this issue in the section “2.3 Metadata” of revised manuscript.

(2) I believe the splitting of the pdb files is a valuable contribution as it standardizes the distribution of antibody-antigen complexes. Indeed, there is great heterogeneity in how many copies of the same structure are present in the structure uploaded to the PDB, generating potential artifacts for machine learning applications to pick up on. That being said, I have two thoughts both for the authors and the broader community. First, in the case of multiple antibodies binding to different epitopes on the same antigen, one should not ignore the potentially stabilizing effect that the binding of one antibody has on the complex, thereby enabling the binding of the second antibody. In general, I urge the community to think about what is the most appropriate spatial context to consider when modeling the stability of interactions from crystal structure data. Second, and in a similar vein, some antigens occur naturally as homomultimers - e.g. influenza hemagglutinin is a homotrimer. Therefore, to analyze the stability of a full-antigen-antibody structure, I believe it would be necessary to consider the full homo-trimer, whereas, in the current curation of AACDB with the proposed data splitting, only the monomers are present.

We sincerely appreciate the reviewer’s insightful comments regarding the splitting of PDB files and we appreciate the opportunity to address the reviewer’s thoughtful concerns.

Firstly, when two antibodies bind to distinct epitopes on the same antigen, we would like to clarify that this scenario can be divided into two cases based on the experimental context: Case1: When two antibodies bind to distinct epitopes on the same antigen, and their complexes are determined in separate structures. For example, SAR650984 (PDB: 4CMH) and daratumumab (PDB: 7DHA) target CD38 at non-overlapping epitopes. These two antibody-antigen complexes were determined independently, and their structures do not influence each other. Case 2 : When the crystal structure contains a ternary complex with two antibodies and an antigen, as in the example of 6OGE discussed in Section 2.2 of our manuscript. After reviewing the original literature, the experiment confirmed that the order of Fab binding does not affect the formation of the ternary complex, and the binding of one antibody does not enhance the binding of the other. This supports the rationale for splitting 6OGE into two separate structures. However, we acknowledge that not all ternary complexes in the PDB provide such detailed experimental descriptions in their original literature. We agree with the reviewer that in some cases, one antibody may stabilize the structure to facilitate the binding of a second antibody. For instance, in 3QUM, the 5D5A5 antibody stabilizes the structure, enabling the binding of the 5D3D11 antibody to human prostate-specific antigen. Such sandwich complexes are indeed valuable for identifying true epitopes and paratopes. Importantly, splitting the structure does not alter the interaction sites.

Secondly, we fully agree with the reviewer that for antigens that naturally exist as homomultimers (e.g., influenza hemagglutinin as a homotrimer), the full multimeric structure should be considered when analyzing stability. In such cases, users can directly utilize the original PDB structures provided in their multimeric form. Our splitting approach is intended to provide an additional option for cases where monomeric analysis is sufficient or preferred, but it does not preclude the use of the original multimeric structures when necessary.

(3) I think the manuscript is lacking in justification about the numbers used as cutoffs ($1\text{\AA}^2$ for change in SASA and 5\AA for maximum distance for contact) The authors just cite other papers applying these two types of cutoffs, but the underlying physico-chemical reasons are not explicit even in these papers. I think that, if the authors want AACDB to be used globally for benchmarks, they should provide direct sources of explanations of the cutoffs used, or provide multiple cutoffs. Indeed, different cutoffs are often used (e.g. ATOM3D uses 6\AA instead of 5\AA to determine contact between a protein and a small molecule <https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/c45147dee729311ef5b5c3003946c48f-Abstract-round1.html>). I think the authors should provide a figure with statistics pertaining to the interface atoms. I think showing any distribution differences between interface atoms determined according to either strategy (number of atoms, correlation between change in SASA and distance...) would be fundamental to understanding the two strategies. I think other statistics would constitute an enhancement as well (e.g. proportion of heavy vs. light chain residues).

Some obvious limitations of AACDB in its current form include:

AACDB only contains entries with protein-based antigens of at most 50 amino acids in length. This excludes non-protein-based antigens, such as carbohydrate- and nucleotide-based, as well as short peptide antigens.

AACDB does not include annotations of binding affinity, which are present in SAbDab and have been proven useful both for characterizing drivers of antibody-antigen interactions (cite <https://www.sciencedirect.com/science/article/pii/S0969212624004362?via%3Dihub>) and for benchmarking antigen-specific antibody-design algorithms (cite <https://www.biorxiv.org/content/10.1101/2023.12.10.570461v1>)).

We thank the reviewer for raising this critical point about the cutoff values used in AACDB. In the current study, the selection of the threshold value is very objective; the threshold chosen in the manuscript is summarized based on existing literature, and we have provided more literature support in the manuscript. The criteria for defining interacting amino acids in established tools, typically do not set the ΔSASA exceed $1\text{\AA}^2$ and the distance exceed 6\AA. While our manuscript emphasizes widely accepted thresholds for consistency with prior benchmarks, AACDB explicitly provides raw ΔSASA and distance values for all annotated residues. Users can dynamically filter the data from downloaded files by excluding entries exceeding their preferred thresholds (e.g., selecting 5\AA instead of 6\AA). This ensures adaptability to diverse research needs. In the revised version, we reset the distance threshold to 6\AA and calculated the interacting amino acids in order to give the user a wider range of choices. In the section “3.2 Database browse and search” of revised manuscript, we provide a description of the flexible choice of thresholds for practical use.

Furthermore, distance and ΔSASA are two distinct metrics for evaluating interactions. Distance directly quantifies spatial proximity between atoms, reflecting physical contacts such as van der Waals interactions or hydrogen bonds, and is ideal for identifying direct spatial adjacency. ΔSASA , on the other hand, measures changes in solvent accessibility of residues during binding, capturing the contribution of buried surfaces to binding free energy. Even for residues not in direct contact, reduced SASA due to conformational changes may indicate indirect functional roles.

As demonstrated through comparisons on the detailed information pages, the sets of interacting amino acids defined by these two methods differ by only a few residues, with no significant variation in their overall distributions. However, since interaction patterns vary significantly across different complexes, analyzing residue distributions across all structures using both criteria is not feasible.

We thank the reviewer for highlighting these limitations. AACDB currently focuses on protein-based antigens ≤ 50 amino acids to prioritize structural consistency, which excludes non-protein antigens and shorter peptides. While affinity annotations are critical for benchmarking antibody design tools, these data were not integrated in this release due to insufficient data verification caused by internal team constraints. We acknowledge these gaps and plan to expand antigen diversity and incorporate affinity metrics in future updates.

Reviewer #2:

Summary:

Antibodies, thanks to their high binding affinity and specificity to cognate protein targets, are increasingly used as research and therapeutic tools. In this work, Zhou et al. have created, curated, and made publicly available a new database of antibody-antigen complexes to support research in the field of antibody modelling, development, and engineering.

Strengths:

The authors have performed a manual curation of antibody-antigen complexes from the Protein Data Bank, rectifying annotation errors; they have added two methods to estimate paratope-epitope interfaces; they have produced a web interface that is capable of both effective visualisation and of summarising the key useful information in one page. The database is also cross-linked to other databases that contain information relevant to antibody developability and therapeutic applications.

Weaknesses:

The database does not import all the experimental information from PDB and contains only complexes with large protein targets.

Thank you for the valuable feedback. As previously responded to Reviewer 1, due to limitations within our team, comprehensive data integration from PDB has not been achieved in the current version. We acknowledge the significance of expanding the database to encompass a broader range of experimental information and complexes with diverse target sizes. Regrettably, immediate updates to address these limitations are not feasible at this time. Nevertheless, we are committed to enhancing the database in upcoming upgrades to provide users with a more comprehensive and inclusive resource

Recommendations for the authors:

Reviewer #1:

(1) Line 194: "produce" → "produced"

We thank the reviewer for the feedback. We have checked the grammar and spelling carefully in the revised manuscript.

(2) As mentioned in the public review, I think adding binding affinity annotations would greatly enhance the use cases for the database.

We thank the reviewer for the suggestion. As the response in "Public review". Due to team constraints, these data are not integrated into this release but are being collated. We recognize these gaps and plan to expand antigenic diversity and incorporate affinity metrics in future updates.

(3) I think adding a visualization of interface atoms and contacts on an entry's webpage would be useful for someone exploring specific entries. It also would be useful if the authors provided a pymol command to select interface residues since that's a procedure any structural biologist is likely to do.

We sincerely appreciate the reviewer's constructive suggestions. In response to the request for enhanced visualization and accessibility of interface residue information, we have implemented the following improvements: (1) Web Interface Visualization. On the entry-specific webpage, we have added an interactive visualization window that highlights the antigen-antibody interaction interface using distinct colors. The interaction interface visualization has been incorporated into Figure 5 of the revised manuscript, with a detailed description. (2) PyMOL Command Accessibility. The "Help" page now provides step-by-step PyMOL commands to select and visualize interface residues.

(4) I think the authors should provide headers to the files containing interface residues according to the change-in-SASA criterion, as they do for those computed according to contact. This would avoid unnecessary confusion - however slight - and make parsing easier. I was initially confused by the meaning of the last column, though after a minute I understood it to be the change in SASA.

We thank the reviewer for providing such detailed feedback. We thank the reviewer for the comment and the suggestion. We have provided headers for the files of the interacting residues defined by Δ SASA.

(5) Line 233: "AACDB's data processing pipeline supports mmCIF files" → The meaning and implications of this statement are not obvious to me, and are mentioned nowhere else in the paper. Do you mean that in AACDB there are structure entries that the RCSB PDB database only has in mmCIF file format, and not .pdb format? So, effectively, there are some entries in AACDB that are not in any other antibody-specific database? I checked and, as of Dec 3rd, 2024, there are 41 structures in AACDB that are NOT in SAbDab. Manually checking 5 of those 41 structures, none are mmCIF-only structures.

We thank the reviewer for the valuable comment. Because of the size of the structures within certain entries, representing them in a single PDB format data file is not feasible due to the excessive number of atoms and polymer chains they contain. As a result, PDB stores these structures in "mmcif" format files. In AACDB, 47 entries, such as 7SOF, 7NKT, 7B27, and 6T9D, are only available in the "mmCIF" format from the PDB. The ".pdb" and ".cif" files contain atomic coordinates in distinct text formats, and the segmentation of these structure files is automatically conducted based on manually annotated antibody-antigen chains. To accommodate this, we have incorporated these considerations into our file processing pipeline, thereby enabling a fully automated file segmentation process. Additionally, we employed Naccess to calculate interatomic distances. However, since this software only accepts .pdb format files as input, we also converted all split .cif files into .pdb format within our fully automated pipeline. We apologize for the lack of clarity in the original manuscript and have included a more detailed explanation in the "2.2 PDB Splitting" section of the revised manuscript.

Reviewer #2:

(1) In SabDab and PDB, experimental binding affinities are also reported: could the authors comment on whether they also imported this information and double-checked it against the original paper? If it wasn't imported, that might discourage some users and should be considered as an extension for the future.

We thank the reviewer for the comment and the suggestion. As the response in “Public review”. Due to current resource constraints, quantitative affinity data has not been incorporated into this release but is undergoing systematic curation. We explicitly recognize these limitations and propose a two-pronged strategy for future iterations: (1) broadening antigen diversity coverage through expanded structural sampling, and (2) integrating quantitative binding affinity measurements. In the Discussion section, we have included description outlining the planned enhancements.

(2) Line 49-50: the references mentioned in connection to deep learning methods for antibody-antigen predictions seem a bit limited given the amount of articles in this field, with 3 of 4 references on one method only (SEPPA), could the authors expand this list to reflect a bit more the state of the art?

We thank the reviewer for the suggestion. We agree that more relevant studies should be listed and therefore more references are provided in the revised manuscript.

When mentioning the limitations of the existing databases, it feels a bit that the criticism is not fully justified. For instance:

Line 52-53: could the authors elaborate on the reasons why such an identification is challenging? (Isn't it possible to make an efficient database-filtered search? Or rather, should one highlight that a more focussed resource is convenient and why?)

Thank you for feedback. In this study, the keywords "antibody complex," "antigen complex," and "immunoglobulin complex," were employed during data collection. PDB returned over 30,000 results, of which only one-tenth met our criteria after rigorous filtering. This demonstrates that keyword searches, while useful, inherently limit result precision and introduce substantial redundancy, likely due to the PDB's search mechanism. That's why we illustrated the significant challenges in identifying antibody-antigen complexes from general protein structures in the PDB.

Line 55: reading the website <http://www.abbybank.org/abdb/>, it would be fairer to say that the web interface lacks updates, as the database and the code have gone through some updates. Could the authors provide a concrete example of the reason why: 'The AbDb database currently lacks proper organization and management of this valuable data.'?

We thank the reviewer for highlighting this issue. In our original manuscript, the statement that the AbDb database "lacks proper organization and management" was based on the absence of explicit statement regarding data updates on its official website at the time of submission, even though internal updates to its content may have occurred. We fully respect the long-standing contributions of AbDb to antibody structural research, and our comments were solely directed at the specific state of the database at that time. As the reviewer noted, following the release of our preprint, we have also taken note of AbDb's recent updates. To reflect the latest developments and avoid potential misinterpretation, we have revised the original statement in revised manuscript.

Also 'this rapid updating process may inadvertently overlook a significant amount of information that requires thorough verification,' it's difficult for me to understand what this means in practice. Could the authors clarify if they simply mean that SabDab collects information from PDB and therefore tends to propagate annotation errors from there? If yes, I think it's enough to state it in these terms, and for sure I agree that the reason is that correcting these annotation errors requires a substantial amount of work.

We thank the reviewer for providing such detailed feedback on the manuscript. We acknowledge that SabDab represents a highly valuable contribution to the field, and its rapid update mechanism has significantly advanced related research areas. However, as stated by the reviewer, we aim to clarify that SabDab primarily relies on automated metadata extraction from the PDB for annotation, and its rapid update process inherently inherits raw data from upstream sources. According to their paper, manual curation is only applied when the automated pipeline fails to resolve structural ambiguities. This workflow—dependent on PDB annotations with limited manual verification—may propagate errors provided by PDB. Examples include species misannotation and mutation status misinterpretation. We fully agree with the reviewer's observation that correcting errors in such cases necessitates labor-intensive manual curation, which is a core motivation for our study.

Line 86: why 'Structures that consisted solely of one type of antibody were excluded'? Why exclude complexes with antigens shorter than 50 amino acids? These complexes are genuine antibody-antigen complexes.

We thank the reviewer for the valuable question. The AACBD database is dedicated to curating structural data of antigen-antibody complexes. Structures featuring only a single antibody type are classified as free antibodies and systematically excluded from the database due to the absence of protein-bound partners. During data screening, we retained sequences shorter than 50 amino acids by categorizing them as peptides rather than eliminating them outright. The current release exclusively encompasses complexes with protein-based antigens. Meanwhile, complexes involving peptide, haptens, and nucleic acid antigens are undergoing systematic curation, with planned inclusion in future updates to broaden antigen category representation.

Line 96 needs a capital letter at the beginning.

Line 107: 'this would generate' → 'this generates' (given it is something that has been implemented, correct?).

Line 124: missing an 'of'.

Line 163: inspiring by -> inspired by.

Thank you for feedback. All of the above grammatical or spelling errors have been revised in the manuscript.

Line 109-111: apart from the example, it would be good to spell out the general rule applied to anti-idiotypic antibodies.

We thank the reviewer for the valuable feedback. For anti-idiotypic antibodies complex, the partner antibody is treated as a dual-chain antigen, necessitating individual evaluation of heavy chain and light chain interactions with the anti-idiotypic component. We have given a general rule for anti-idiotypic antibodies in section “2.2 PDB splitting” of revised manuscript.

Line 155-159: could the authors provide references for the two choices (based on sasa and any-atom distance) that they adopted to define interacting residues?

We thank the reviewer for the comment and the suggestion. As the same as the response to reviewer #1 in Public review. The interacting residues definition and the threshold chosen in the manuscript is summarized based on existing literature. We have added additional references for support in section “1.Introduction”. Our resource does not provide a fixed amino acid list. Instead, all interacting residues are explicitly documented alongside their

corresponding Δ SASA (solvent-accessible surface area changes) and intermolecular distances, allowing researchers to flexibly select residue pairs based on customized thresholds from downloadable datasets. Furthermore, aligning with widely adopted criteria in current literature—where interactions are defined by Δ SASA $>1 \text{ \AA}^2$ and atomic distances $<6 \text{ \AA}$, we have recalibrated our analysis in the revised version. Specifically, we replaced the previous $5 \text{ \AA}$ distance threshold with a $6 \text{ \AA}$ cutoff to recalculate interacting residues.

Line 176-178: could the authors re-phrase this sentence to clarify what they mean by 'change in the distribution'?

We thank the reviewer for the suggestion. Our search was conducted with an end date of November 2023. However, Figure 3B includes an entry dated 2024. Upon reviewing this record, we identified that the discrepancy arises from the supersession of the 7SIX database entry (originally released in December 2022) by the 8TM1 version in January 2024. This version update explains the apparent chronological inconsistency. We regret any lack of clarity in our original description and have revised the corresponding section in the manuscript to explicitly clarify this change of database.

Caption Figure 3: please spell out all the acronyms in the figure. Provide the date when the last search was performed (i.e., the date of the last update of these statistics).

We thank the reviewer for the comment. We have systematically expanded all acronyms and included update dates for statistics in the legend of Figure 3. Corresponding changes have also been made to the statistical pages on the website.

Finally, it would be advisable to do a general check on the use of the English language (e.g. I noted a few missing articles). In Figure 5 DrugBank contains typos.

We sincerely appreciate the reviewer's meticulous attention to linguistic precision. We have corrected the typographical error in Figure 5 and conducted a comprehensive review of the entire manuscript to ensure accuracy and clarity.

<https://doi.org/10.7554/eLife.104934.2.sa0>